

Table 3. As can be seen, the 3D enhanced tubes have the highest internal Wilson plot coefficient and highest internal heat transfer due to the advanced helical structures present on the inner tube surface.

2.5.4. Thermal wall resistance

The resistance to heat transfer the wall poses was calculated through equation (5):

$$R_t = \frac{D_o \cdot \ln\left(\frac{D_o}{D_i}\right)}{2 \cdot k_t} \quad (5)$$

Where k_t is the estimated thermal conductivity of copper, and was taken as k_t of 340 W/mK [34].

2.5.5. Outside tube surface heat transfer coefficient

Considering the thermal network and when normalized to outside tube surface area, the external heat transfer coefficient, h_o , is given by equation (6):

$$h_o = \left(\frac{1}{U_{o,mid}} - R_t - \frac{1}{h_i} \cdot \frac{D_o}{D_i} \right)^{-1} \quad (6)$$

Where $U_{o,mid}$ is the determined overall heat transfer coefficient with respect to the middle of the tube, R_t the calculated thermal wall resistance, h_i the determined internal heat transfer coefficient and D_o and D_i the respective outer and inner diameter of the tube.

2.5.6. Heat flux exponent

The influence of heat flux are quantified by equation (7):

$$h_o = a \cdot \dot{q}^{n_q} \quad (7)$$

2.5.7. Enhancement heat transfer influence ratio

The effect of the 3D surface enhancements on heat transfer were observed through a ratio of the heat transfer coefficients of the micro-structured tube to those of the plain roughened tube at the same testing heat flux point. This was done by calculating a polynomial best-fit curve to each of the heat transfer curves and dividing the HTC of each of the micro-enhanced tubes by the HTCs of the plain roughened tube over the testing heat flux range in intervals of 10 kW/m². This ratio, K_{surf} , was